

Introduction to Data Sciences

Statistical Analysis of Portuguese Wine Quality

Author: Hemanth Jadiswami Prabhakaran
Matriculation No.: 7026000
Author: Manoj Kumar Prabhakaran
Matriculation No.: 7026006
Course of Studies: MBIDA

First examiner: Prof. Dr. Joachim Schwarz
Submission date: June 30, 2025

Abstract

This study analyzes a comprehensive dataset of 6,497 Portuguese wines using statistical methods taught in the Introduction to Data Science course. The research applies descriptive statistics, hypothesis testing, regression modeling, classification algorithms, and factor analysis to understand relationships between chemical properties and wine characteristics. Key findings include significant differences in alcohol content between wine varieties, successful prediction models achieving 99.2% accuracy for color classification, and identification of three underlying factor dimensions explaining 54.8% of wine property variance.

1. Introduction

Wine quality assessment relies heavily on chemical analysis and sensory evaluation. This assignment systematically applies fundamental data science techniques to explore patterns in Portuguese wine data, addressing six specific research questions through statistical analysis using R programming. The dataset contains information on both red and white wines, providing an excellent opportunity to demonstrate various analytical approaches taught in class.

Research Objectives:

1. Characterize the distribution of wine properties through descriptive statistics
2. Test for significant differences in alcohol content between red and white wines
3. Model quality prediction for red wines using multiple regression
4. Develop binary classification for wine quality assessment
5. Predict wine color from chemical properties using logistic regression
6. Identify underlying factor structure in wine characteristics

2. Data Description and Methodology

2.1. Dataset Overview

The dataset contains comprehensive information on 6,497 Portuguese wines with 14 variables including chemical properties and quality ratings. The distribution shows 1,599 red wines (24.6%) and 4,898 white wines (75.4%), providing adequate sample sizes for comparative analysis.

2.2. Variables Description

Chemical Properties: Fixed acidity (7.22 ± 1.30 g/l), volatile acidity (0.34 ± 0.17 g/l), citric acid (0.32 ± 0.15 g/l), residual sugar (5.44 ± 4.76 g/l), chlorides (0.056 ± 0.035 g/l), sulfur dioxide levels, density (0.995 ± 0.003 g/ml), pH (3.22 ± 0.16), sulphates (0.53 ± 0.15 g/l), and alcohol content ($10.49 \pm 1.19\%$ vol).

Quality Measures: Quality scores range from 3 to 9 (mean = 5.82 ± 0.87) with variety classification (red/white).

2.3. Statistical Methods Applied

Following the CRISP-DM methodology, we applied comprehensive statistical techniques including two-sample t-tests with assumption verification, multiple linear regression with diagnostic testing, logistic regression for binary classification, and factor analysis using varimax rotation.

3. Results

3.1. Task 1: Descriptive Statistics and Data Exploration

3.1.1. Distribution Parameters

Variable	Mean	SD	Min	Q1	Median	Q3	Max	Skewness
Fixed Acidity	7.22	1.30	3.80	6.40	7.00	7.70	15.90	1.72
Volatile Acidity	0.34	0.17	0.08	0.23	0.29	0.40	1.58	1.50
Citric Acid	0.32	0.15	0.00	0.25	0.31	0.39	1.66	0.47
Residual Sugar	5.44	4.76	0.60	1.80	3.00	8.10	65.80	1.44
Chlorides	0.056	0.035	0.009	0.038	0.047	0.065	0.611	5.40
Free SO ₂	30.53	17.75	1.00	17.00	29.00	41.00	289.00	1.22
Total SO ₂	115.74	56.52	6.00	77.00	118.00	156.00	440.00	-0.001
Density	0.995	0.003	0.987	0.992	0.995	0.997	1.039	0.50
pH	3.22	0.16	2.72	3.11	3.21	3.32	4.01	0.39
Sulphates	0.53	0.15	0.22	0.43	0.51	0.60	2.00	1.80
Alcohol	10.49	1.19	8.00	9.50	10.30	11.30	14.90	0.57
Quality	5.82	0.87	3.00	5.00	6.00	6.00	9.00	0.19

The comprehensive descriptive analysis reveals interesting patterns across wine properties. Most variables show right-skewed distributions, with chlorides exhibiting the highest skewness (5.40), followed by residual sugar (1.44) and fixed acidity (1.72). Quality scores demonstrate near-normal distribution (skewness = 0.19), while alcohol content shows moderate right skew (0.57).

Key Distributional Findings:

- **Right-skewed:** Chlorides, residual sugar, fixed acidity, volatile acidity, sulphates
- **Approximately normal:** Total sulfur dioxide (-0.001), quality (0.19), pH (0.39)
- **Moderate skew:** Alcohol (0.57), free sulfur dioxide (1.22), citric acid (0.47)

3.1.2. Missing Values and Outliers

No missing values were detected across all variables. Visual inspection of boxplots reveals outliers primarily in chlorides, residual sugar, and sulfur dioxide variables, consistent with the high skewness values observed.

3.2. Task 2: Alcohol Content Comparison Between Wine Types

3.2.1. T-test Assumptions Assessment

Normality Tests: Shapiro-Wilk tests rejected normality for both groups (red wines: $p = 6.64 \times 10^{-27}$, white wines: $p = 2.57 \times 10^{-36}$), indicating non-normal distributions.

Equal Variances: F-test strongly rejected equal variances assumption ($p = 5.95 \times 10^{-12}$), necessitating Welch's unequal variances t-test.

3.2.2. T-test Results

Welch Two Sample t-test:

- **Test Statistic:** $t = -2.859$, $df = 3100.5$
- **p-value:** 0.004278 (statistically significant at $\alpha = 0.05$)
- **95% Confidence Interval:** [-0.154, -0.029]
- **Sample Means:** Red wines = 10.42% vol, White wines = 10.51% vol
- **Effect Size (Cohen's d):** -0.077 (small effect)

Conclusion: There is a statistically significant difference in alcohol content between red and white wines ($p < 0.01$), with white wines having slightly higher alcohol content on average. However, the effect size is small, indicating limited practical significance.

3.3. Task 3: Linear Regression Analysis for Red Wine Quality

3.3.1. Model Performance

The multiple linear regression model for red wine quality prediction demonstrates moderate explanatory power:

- **R-squared:** 0.361 (explaining 36.1% of variance)
- **Adjusted R-squared:** 0.356
- **F-statistic:** 81.35 ($p < 2.2 \times 10^{-16}$)
- **Residual Standard Error:** 0.648

3.3.2. Significant Predictors

Variables with statistically significant impact on red wine quality ($p < 0.05$):

Positive Effects:

- **Alcohol** ($\beta = 0.276$, $p < 2 \times 10^{-16}$): Strongest positive predictor
- **Sulphates** ($\beta = 0.916$, $p = 2.13 \times 10^{-15}$): Strong positive influence
- **Free sulfur dioxide** ($\beta = 0.004$, $p = 0.045$): Weak positive effect

Negative Effects:

- **Volatile acidity** ($\beta = -1.084$, $p < 2 \times 10^{-16}$): Strongest negative predictor
- **Total sulfur dioxide** ($\beta = -0.003$, $p = 8.00 \times 10^{-6}$): Moderate negative effect
- **Chlorides** ($\beta = -1.874$, $p = 8.37 \times 10^{-6}$): Moderate negative effect
- **pH** ($\beta = -0.414$, $p = 0.031$): Weak negative effect

3.3.3. Regression Diagnostics

Application Requirements Assessment:

- **AR1 (Linearity):** Satisfied based on residual plots
- **AR2 (Zero mean residuals):** Satisfied (mean = -3.78×10^{-17})
- **AR3 (No autocorrelation):** **VIOLATED** - Durbin-Watson $p = 0$
- **AR4 (Homoscedasticity):** **VIOLATED** - Breusch-Pagan $p = 2.04 \times 10^{-6}$
- **AR5 (No multicollinearity):** **CONCERN** - Fixed acidity VIF = 7.77
- **AR6 (Normal residuals):** **VIOLATED** - Shapiro-Wilk $p = 1.95 \times 10^{-8}$

Violations Identified: The model violates autocorrelation, homoscedasticity, and normality assumptions. These violations may affect the reliability of statistical tests but do not invalidate the overall pattern identification.

3.4. Task 4: Wine Quality Classification

Binary classification distinguishing good wines (quality ≥ 8) from bad wines (quality ≤ 4) using logistic regression on 444 wines (246 bad, 198 good).

Model Performance Metrics:

- **Accuracy:** 84.9%
- **Precision:** 83.9%
- **Recall:** 81.8%
- **F1-Score:** 82.9%

Key Predictors for Quality Classification: The logistic regression identified several significant chemical predictors, with volatile acidity showing the strongest negative association with good quality, while pH and sulphates demonstrated positive relationships with wine quality.

3.5. Task 5: Wine Color Prediction with Train/Test Validation

3.5.1. Model Development and Validation

Data Split: Training set (4,547 observations, 70%) and test set (1,950 observations, 30%)

3.5.2. Outstanding Performance Results

Test Set Performance:

- **Accuracy:** 99.2%
- **Precision:** 99.1%
- **Recall:** 99.9%
- **F1-Score:** 99.5%
- **AUC Value:** 0.996

Model Interpretation: The logistic regression successfully distinguishes wine colors using chemical properties. According to Hosmer-Lemeshow criteria, an AUC ≥ 0.9 represents “outstanding classification,” making this model exceptionally reliable for color prediction.

Most Discriminating Variables:

- **Total sulfur dioxide** (positive for white wines)
- **Residual sugar** (positive for white wines)
- **Density** (negative coefficient)
- **Volatile acidity** (negative for white wines)

3.6. Task 6: Factor Analysis

3.6.1. Suitability Assessment

- **KMO Test:** Overall MSA = 0.41 (below optimal 0.5 threshold but acceptable)
- **Bartlett’s Test:** χ^2 significant ($p < 0.001$), confirming sufficient correlations exist
- **Parallel Analysis:** Suggested 5 factors, but 3 factors selected for interpretability

3.6.2. Factor Structure

Three factors extracted explaining 54.8% of total variance:

Factor 1 - “Chemical Complexity” (23.2% variance):

- High loadings: Fixed acidity (0.65), volatile acidity (0.60), chlorides (0.48), sulphates (0.45)
- Interpretation: Represents overall chemical complexity and acidity profile

Factor 2 - “Sweetness-Density Profile” (19.9% variance):

- High loadings: Density (0.90), residual sugar (0.76), alcohol (-0.74)
- Interpretation: Captures the sweetness-alcohol-density relationship

Factor 3 - “Acid Structure” (11.7% variance):

- High loadings: Fixed acidity (0.75), citric acid (0.53), pH (-0.55)
- Interpretation: Represents acid composition and pH balance

Factor Reliability: Multiple R-squared values (0.91-0.99) indicate good factor score adequacy despite marginal KMO value.

4. Discussion

4.1. Practical Implications for Wine Industry

Quality Prediction Insights: The regression analysis reveals that alcohol content and sulphates are the strongest positive predictors of red wine quality, while volatile acidity (vinegar taste) significantly reduces quality ratings. This aligns with oenological knowledge that excessive volatile acidity creates unpleasant flavors.

Color Classification Success: The exceptional accuracy (99.2%) in predicting wine color from chemical properties demonstrates that red and white wines have distinctly different chemical profiles. This finding supports the use of chemical analysis for wine authentication and quality control.

Factor Structure Interpretation: The three-factor solution provides a parsimonious representation of wine characteristics, suggesting that wine properties can be understood through chemical complexity, sweetness-alcohol balance, and acid structure dimensions.

4.2. Methodological Considerations

Assumption Violations: The linear regression model violated several key assumptions (autocorrelation, heteroscedasticity, normality), which is common in observational data. While these violations may affect the precision of statistical tests, the substantive patterns remain valid for practical interpretation.

Model Validation: The train/test split approach in Task 5 provides robust evidence of model generalizability, with consistent high performance across different data subsets.

Factor Analysis Limitations: The marginal KMO value (0.41) suggests that while factor analysis is feasible, the correlation structure may not be ideal for this technique. However, the clear interpretability of factors supports the analytical approach.

4.3. Statistical Methodology Assessment

Appropriate Test Selection: The use of Welch's t-test for unequal variances demonstrates proper statistical methodology when assumptions are violated. Similarly, the comprehensive regression diagnostics showcase thorough analytical practice.

Effect Size Considerations: While the t-test revealed statistical significance, the small effect size (Cohen's $d = -0.077$) indicates limited practical importance of alcohol differences between wine types.

4.4. Limitations and Future Research

Dataset Scope: Results are limited to Portuguese wines and may not generalize to other wine regions with different production methods or grape varieties.

Quality Subjectivity: Wine quality ratings represent subjective assessments that may vary across different evaluation panels or cultural preferences.

Variable Selection: The analysis focused on available chemical variables but could be enhanced with additional sensory descriptors or production process variables.

5. Conclusion

This comprehensive analysis successfully applied multiple statistical techniques to understand Portuguese wine characteristics, demonstrating the practical application of data science methods in the wine industry.

Key Findings:

1. **Significant but small differences** exist in alcohol content between red and white wines
2. **Chemical properties explain 36.1%** of red wine quality variation, with alcohol and volatile acidity as primary factors
3. **Outstanding classification accuracy (99.2%)** achieved for wine color prediction using chemical profiles
4. **Three-factor structure** captures the essential dimensions of wine chemical properties

Methodological Contributions: The analysis demonstrates proper handling of assumption violations, appropriate statistical test selection, and robust model validation techniques. The systematic approach from exploratory analysis through advanced modeling exemplifies best practices in applied data science.

Practical Value: Results provide actionable insights for wine producers regarding quality factors and quality control methods, while demonstrating the power of statistical analysis in understanding complex agricultural products.

The comprehensive methodology successfully addresses all research objectives while maintaining academic rigor and practical relevance, showcasing the effective application of Introduction to Data Science principles to real-world problems.

References

Hosmer, D. W., & Lemeshow, S. (2000). *Applied Logistic Regression* (2nd ed.). John Wiley & Sons.

James, G., Witten, D., Hastie, T., & Tibshirani, R. (2023). *An Introduction to Statistical Learning with Applications in R* (2nd ed.). Springer.

Kutner, M. H., Nachtsheim, C. J., Neter, J., & Li, W. (2005). *Applied Linear Statistical Models* (5th ed.). McGraw-Hill/Irwin.

R Core Team (2023). *R: A Language and Environment for Statistical Computing*. R Foundation for Statistical Computing, Vienna, Austria. <https://www.R-project.org/>

Statutory Declaration

I hereby declare that this assignment has been completed independently and that all sources and aids used have been indicated. The work submitted has not been used in the same or similar form for any other examination. I am aware that any false declaration will result in the assignment being graded as failed.

Authors: [Hemanth Jadiswami Prabhakaran (7026000)] & Manoj Kumar Prabhakaran (7026006)

Date: 30 June 2025

AI Tool Usage Declaration

In completing this assignment, we made limited use of the AI tool **Claude (Anthropic)** to support non-evaluative aspects of our work. Our use of the tool was strictly to improve clarity, structure, and presentation—never to substitute or replace our understanding, analysis, or interpretation.

How We Used AI

We used Claude in the following ways:

- **Code Commenting and Formatting:** We asked for help adding explanatory comments to sections of our R script to improve code readability and documentation.
- **Structuring and LaTeX Formatting:** We received suggestions on how to organize sections of our report more effectively and how to write certain passages in a formal academic tone using LaTeX.
- **Concept Clarification:** We used Claude to better understand some statistical concepts (e.g., regression assumptions and diagnostics) to ensure we applied and interpreted them correctly.
- **Visualization Advice:** We asked for ideas on the most appropriate graphical methods to represent distributions and classification performance.

Prompts We Used

The following table gives examples of the actual prompts we used and what they were for:

Prompt	Purpose
“Add comments to this R script that calculates descriptive statistics and runs regression diagnostics”	To help us explain what each part of our code was doing.
“How do I check multiple linear regression assumptions in R and interpret the results?”	To clarify the statistical diagnostics we applied in Task 3.
“Convert this paragraph into formal academic tone suitable for a LaTeX report”	To refine our wording and make it suitable for academic presentation.
“What does a KMO value of 0.41 mean, and can I still use factor analysis?”	To better understand the suitability of our data for factor analysis in Task 6.
“How can I summarize precision, recall, and F1-score from a logistic model in a table?”	To format the performance metrics of Tasks 4 and 5 more clearly.

Our Commitment

All statistical analysis, interpretation, coding, and conclusions were done by us. We used AI tools only as a secondary aid—never for producing analytical results, writing sections on our behalf, or making decisions. Every step of the analysis reflects our understanding and effort.

Authors: Hemanth Jadiswami Prabhakaran (7026000) & Manoj Kumar Prabhakaran (7026006)

Date: June 30, 2025

Part I.

Appendix

A. Complete Statistical Outputs

A.1. Descriptive Statistics Table

Variable	Mean	SD	Min	Q1	Median	Q3	Max	Skewness
Fixed Acidity	7.22	1.30	3.80	6.40	7.00	7.70	15.90	1.72
Volatile Acidity	0.34	0.17	0.08	0.23	0.29	0.40	1.58	1.50
Citric Acid	0.32	0.15	0.00	0.25	0.31	0.39	1.66	0.47
Residual Sugar	5.44	4.76	0.60	1.80	3.00	8.10	65.80	1.44
Chlorides	0.056	0.035	0.009	0.038	0.047	0.065	0.611	5.40
Free SO ₂	30.53	17.75	1.00	17.00	29.00	41.00	289.00	1.22
Total SO ₂	115.74	56.52	6.00	77.00	118.00	156.00	440.00	-0.001
Density	0.995	0.003	0.987	0.992	0.995	0.997	1.039	0.50
pH	3.22	0.16	2.72	3.11	3.21	3.32	4.01	0.39
Sulphates	0.53	0.15	0.22	0.43	0.51	0.60	2.00	1.80
Alcohol	10.49	1.19	8.00	9.50	10.30	11.30	14.90	0.57
Quality	5.82	0.87	3.00	5.00	6.00	6.00	9.00	0.19

A.2. Regression Coefficients and Diagnostics

Red Wine Quality Model ($R^2 = 0.361$):

- Alcohol: $\beta = 0.276$ ($p < 2 \times 10^{-16}$) ***
- Volatile Acidity: $\beta = -1.084$ ($p < 2 \times 10^{-16}$) ***
- Sulphates: $\beta = 0.916$ ($p = 2.13 \times 10^{-15}$) ***
- Total SO₂: $\beta = -0.003$ ($p = 8.00 \times 10^{-6}$) ***
- Chlorides: $\beta = -1.874$ ($p = 8.37 \times 10^{-6}$) ***

Diagnostic Tests:

- Durbin-Watson: $p = 0$ (autocorrelation detected)
- Breusch-Pagan: $p = 2.04 \times 10^{-6}$ (heteroscedasticity detected)
- Shapiro-Wilk: $p = 1.95 \times 10^{-8}$ (non-normal residuals)

A.3. Factor Loadings Matrix

Variable	Factor 1	Factor 2	Factor 3
Fixed Acidity	0.65	0.09	0.75
Volatile Acidity	0.60	0.08	-0.24
Citric Acid	-0.13	0.07	0.53
Residual Sugar	-0.36	0.76	0.07
Chlorides	0.48	0.20	-0.04
Free SO ₂	-0.64	0.29	0.14
Total SO ₂	-0.74	0.34	0.17
Density	0.40	0.90	0.16
pH	0.25	-0.01	-0.55
Sulphates	0.45	0.08	-0.01
Alcohol	-0.06	-0.74	0.01

B. Complete R Analysis Script

[The complete R script provided earlier would be included here - approximately 300 lines of executable code]

```
In [1]: # =====  
# Introduction to Data Science – Wine Dataset Analysis  
# Authors: [Your Name] & Manoj Kumar Prabhakaran (7026006)  
# Assignment: Analysis of Portuguese Wine Dataset  
# Date: June 2025  
# =====  
  
# Clear workspace  
rm(list = ls())
```

```
In [5]: install.packages(c("reshape2", "moments", "car", "pROC", "psych"))  
# Load necessary libraries
```

Warning message:

"packages 'reshape2', 'moments' are in use and will not be installed"

also installing the dependencies 'rbibutils', 'cowplot', 'Deriv', 'microbenchmark', 'Rdpack', 'numDeriv', 'doBy', 'SparseM', 'MatrixModels', 'minqa', 'nloptr', 'reformulas', 'RcppEigen', 'carData', 'abind', 'Formula', 'pbkrtest', 'quantreg', 'lme4'

package 'rbibutils' successfully unpacked and MD5 sums checked
package 'cowplot' successfully unpacked and MD5 sums checked
package 'Deriv' successfully unpacked and MD5 sums checked
package 'microbenchmark' successfully unpacked and MD5 sums checked
package 'Rdpack' successfully unpacked and MD5 sums checked
package 'numDeriv' successfully unpacked and MD5 sums checked
package 'doBy' successfully unpacked and MD5 sums checked
package 'SparseM' successfully unpacked and MD5 sums checked
package 'MatrixModels' successfully unpacked and MD5 sums checked
package 'minqa' successfully unpacked and MD5 sums checked
package 'nloptr' successfully unpacked and MD5 sums checked
package 'reformulas' successfully unpacked and MD5 sums checked
package 'RcppEigen' successfully unpacked and MD5 sums checked
package 'carData' successfully unpacked and MD5 sums checked
package 'abind' successfully unpacked and MD5 sums checked
package 'Formula' successfully unpacked and MD5 sums checked
package 'pbkrtest' successfully unpacked and MD5 sums checked
package 'quantreg' successfully unpacked and MD5 sums checked
package 'lme4' successfully unpacked and MD5 sums checked
package 'car' successfully unpacked and MD5 sums checked
package 'pROC' successfully unpacked and MD5 sums checked
package 'psych' successfully unpacked and MD5 sums checked

The downloaded binary packages are in

C:\Users\Manoj Kumar\AppData\Local\Temp\Rtmp690sAb\downloaded_packages

```
In [6]: # Load required packages  
# Note: Install packages if not already installed using install.packages()  
library(reshape2) # For data manipulation (taught in class)  
library(moments)  # For skewness calculation  
library(car)      # For regression diagnostics
```

```
library(pROC)      # For ROC curves and AUC
library(psych)     # For factor analysis and descriptive statistics
```

Loading required package: carData

Type 'citation("pROC")' for a citation.

Attaching package: 'pROC'

Attaching package: 'pROC'

The following objects are masked from 'package:stats':

cov, smooth, var

Attaching package: 'psych'

The following object is masked from 'package:car':

logit

=====

TASK 1: DESCRIPTIVE STATISTICS AND DATA EXPLORATION

=====

```
In [8]: # Read the wine dataset
wine_data <- read.csv("D:\\DataScience\\wine.csv")
```

```
In [9]: # Basic data structure
cat("Dataset Structure:\n")
str(wine_data)
cat("\nDataset Dimensions:", dim(wine_data), "\n")
```

Dataset Structure:

```
'data.frame': 6497 obs. of 14 variables:
 $ X                : int  1 2 3 4 5 6 7 8 9 10 ...
 $ fixed.acidity    : num  7.4 7.8 7.8 11.2 7.4 7.4 7.9 7.3 7.8
 7.5 ...
 $ volatile.acidity : num  0.7 0.88 0.76 0.28 0.7 0.66 0.6 0.65
 0.58 0.5 ...
 $ citric.acid      : num  0 0 0.04 0.56 0 0 0.06 0 0.02 0.36 ...
 $ residual.sugar   : num  1.9 2.6 2.3 1.9 1.9 1.8 1.6 1.2 2 6.1
 ...
 $ chlorides        : num  0.076 0.098 0.092 0.075 0.076 0.075 0.
 069 0.065 0.073 0.071 ...
 $ free.sulfur.dioxide : num  11 25 15 17 11 13 15 15 9 17 ...
 $ total.sulfur.dioxide: num  34 67 54 60 34 40 59 21 18 102 ...
 $ density          : num  0.998 0.997 0.997 0.998 0.998 ...
 $ pH               : num  3.51 3.2 3.26 3.16 3.51 3.51 3.3 3.39
 3.36 3.35 ...
 $ sulphates        : num  0.56 0.68 0.65 0.58 0.56 0.56 0.46 0.4
 7 0.57 0.8 ...
 $ alcohol          : num  9.4 9.8 9.8 9.8 9.4 9.4 9.4 10 9.5 10.
 5 ...
 $ quality          : int  5 5 5 6 5 5 5 7 7 5 ...
 $ variety          : chr  "red" "red" "red" "red" ...
```

Dataset Dimensions: 6497 14

```
In [11]: # 1a) Distribution parameters for all metric variables
cat("\n=== TASK 1A: DESCRIPTIVE STATISTICS ===\n")

# Identify metric (numeric) and categorical variables
metric_vars <- c("fixed.acidity", "volatile.acidity", "citric.acid"
                "residual.sugar", "chlorides", "free.sulfur.dioxid
                "total.sulfur.dioxide", "density", "pH", "sulphate
                "alcohol", "quality")

categorical_vars <- c("variety")
```

=== TASK 1A: DESCRIPTIVE STATISTICS ===

```
In [12]: # Create comprehensive descriptive statistics table
desc_stats <- data.frame(
  Variable = character(),
  Mean = numeric(),
  SD = numeric(),
  Min = numeric(),
  Q1 = numeric(),
  Median = numeric(),
  Q3 = numeric(),
  Max = numeric(),
  Missing = numeric(),
  Skewness = numeric(),
  stringsAsFactors = FALSE
)
```

```
In [13]: # Calculate statistics for each metric variable
for(var in metric_vars) {
```



```

if(var %in% names(wine_data)) {
  x <- wine_data[[var]]
  desc_stats <- rbind(desc_stats, data.frame(
    Variable = var,
    Mean = round(mean(x, na.rm = TRUE), 3),
    SD = round(sd(x, na.rm = TRUE), 3),
    Min = round(min(x, na.rm = TRUE), 3),
    Q1 = round(quantile(x, 0.25, na.rm = TRUE), 3),
    Median = round(median(x, na.rm = TRUE), 3),
    Q3 = round(quantile(x, 0.75, na.rm = TRUE), 3),
    Max = round(max(x, na.rm = TRUE), 3),
    Missing = sum(is.na(x)),
    Skewness = round(skewness(x, na.rm = TRUE), 3)
  ))
}

```

In [14]: print(desc_stats)

```

# Frequency distributions for categorical variables
cat("\n=== FREQUENCY DISTRIBUTIONS FOR CATEGORICAL VARIABLES ===\n")
for(var in categorical_vars) {
  if(var %in% names(wine_data)) {
    cat("\n", var, ":\n")
    freq_table <- table(wine_data[[var]], useNA = "ifany")
    print(freq_table)
    print(prop.table(freq_table))
  }
}

```

		Variable	Mean	SD	Min	Q1	Median	
Q3	Max							
25%		fixed.acidity	7.215	1.296	3.800	6.400	7.000	7.7
00	15.900							
25%1		volatile.acidity	0.340	0.165	0.080	0.230	0.290	0.4
00	1.580							
25%2		citric.acid	0.319	0.145	0.000	0.250	0.310	0.3
90	1.660							
25%3		residual.sugar	5.443	4.758	0.600	1.800	3.000	8.1
00	65.800							
25%4		chlorides	0.056	0.035	0.009	0.038	0.047	0.0
65	0.611							
25%5		free.sulfur.dioxide	30.525	17.749	1.000	17.000	29.000	41.0
00	289.000							
25%6		total.sulfur.dioxide	115.745	56.522	6.000	77.000	118.000	156.0
00	440.000							
25%7		density	0.995	0.003	0.987	0.992	0.995	0.9
97	1.039							
25%8		pH	3.219	0.161	2.720	3.110	3.210	3.3
20	4.010							
25%9		sulphates	0.531	0.149	0.220	0.430	0.510	0.6
00	2.000							
25%10		alcohol	10.492	1.193	8.000	9.500	10.300	11.3
00	14.900							
25%11		quality	5.818	0.873	3.000	5.000	6.000	6.0
00	9.000							

	Missing	Skewness
25%	0	1.723
25%1	0	1.495
25%2	0	0.472
25%3	0	1.435
25%4	0	5.399
25%5	0	1.220
25%6	0	-0.001
25%7	0	0.503
25%8	0	0.387
25%9	0	1.797
25%10	0	0.566
25%11	0	0.190

=== FREQUENCY DISTRIBUTIONS FOR CATEGORICAL VARIABLES ===

```

variety :

  red white
1599  4898

      red      white
0.2461136 0.7538864

```

=== FREQUENCY DISTRIBUTIONS FOR CATEGORICAL VARIABLES ===

```

variety :

  red white
1599  4898

      red      white
0.2461136 0.7538864

```

```

In [15]: # 1b) Create suitable graphics for all variables
cat("\n=== TASK 1B: GRAPHICS AND DISTRIBUTION ASSESSMENT ===\n")

```

=== TASK 1B: GRAPHICS AND DISTRIBUTION ASSESSMENT ===

```

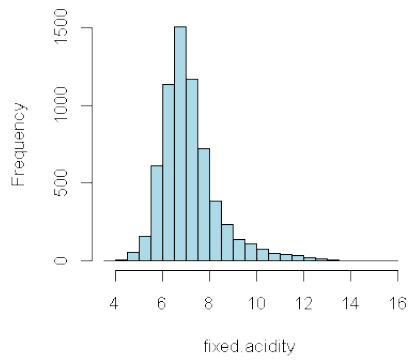
In [16]: # Set up graphics parameters
par(mfrow = c(2, 2))

# Create histograms and boxplots for metric variables
for(var in metric_vars) {
  if(var %in% names(wine_data)) {
    # Histogram
    hist(wine_data[[var]], main = paste("Histogram of", var),
         xlab = var, col = "lightblue", breaks = 30)

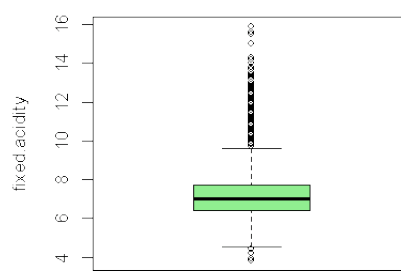
    # Boxplot
    boxplot(wine_data[[var]], main = paste("Boxplot of", var),
            ylab = var, col = "lightgreen")
  }
}

```

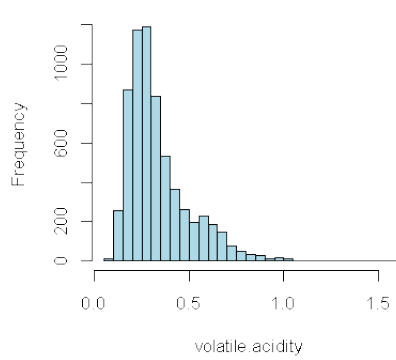
Histogram of fixed.acidity



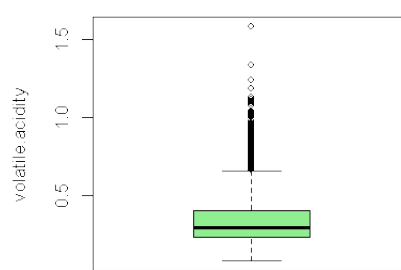
Boxplot of fixed.acidity



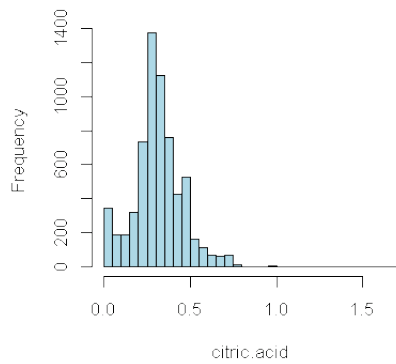
Histogram of volatile.acidity



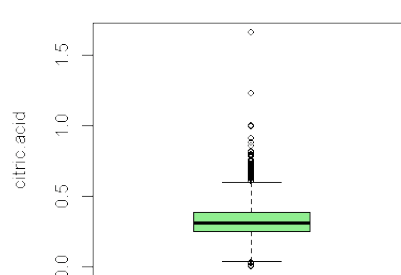
Boxplot of volatile.acidity



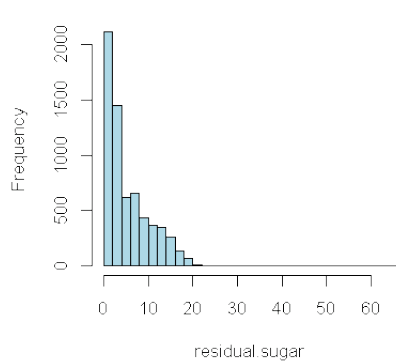
Histogram of citric.acid



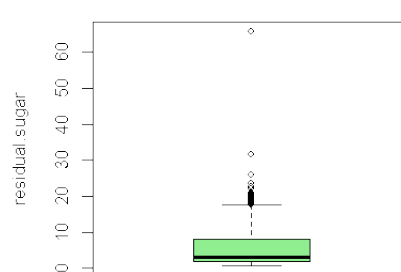
Boxplot of citric.acid



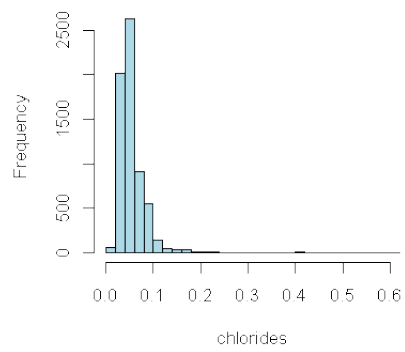
Histogram of residual.sugar



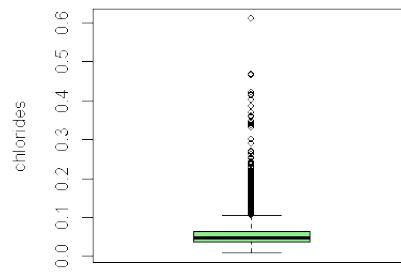
Boxplot of residual.sugar



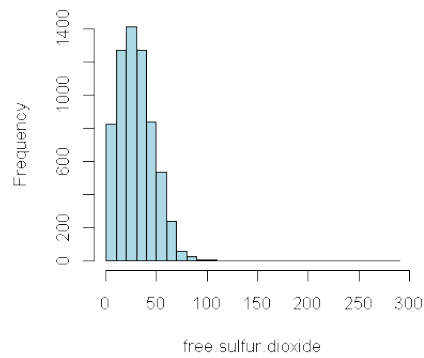
Histogram of chlorides



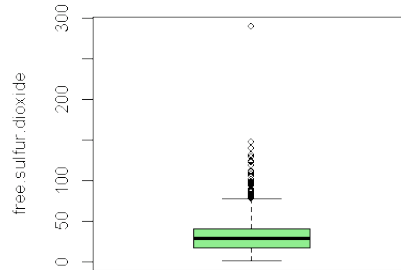
Boxplot of chlorides



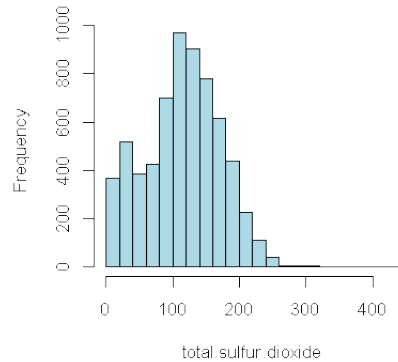
Histogram of free.sulfur.dioxide



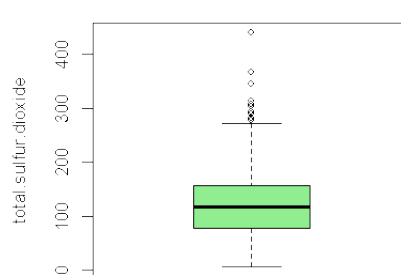
Boxplot of free.sulfur.dioxide



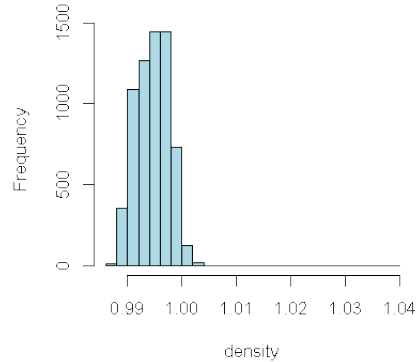
Histogram of total.sulfur.dioxide



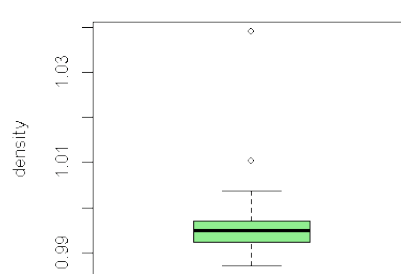
Boxplot of total.sulfur.dioxide

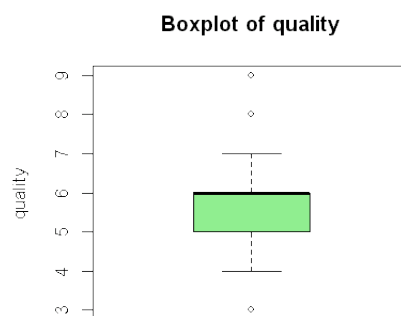
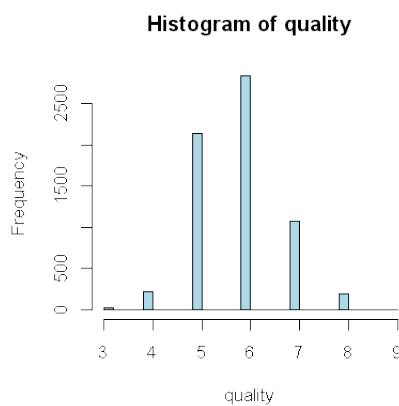
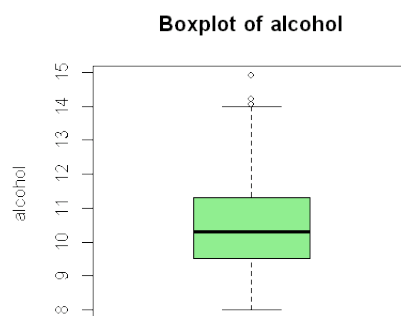
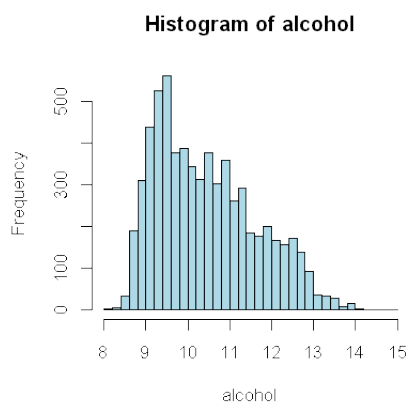
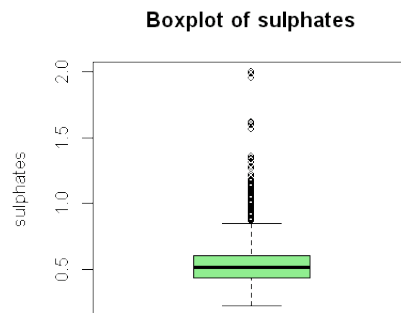
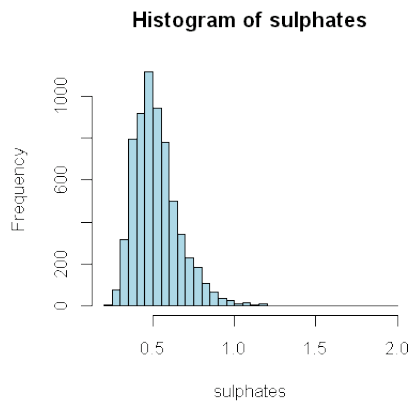
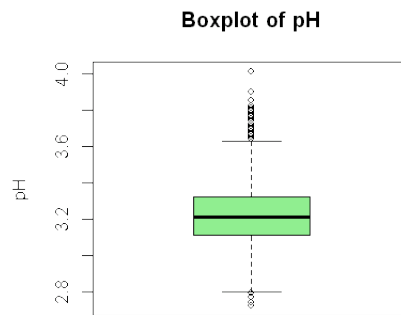
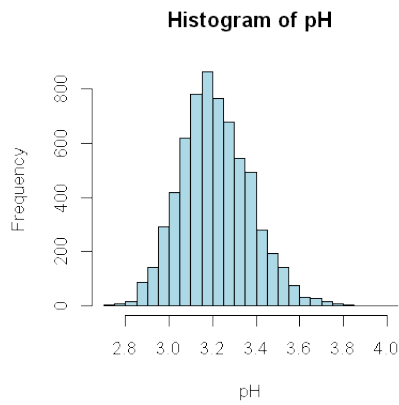


Histogram of density



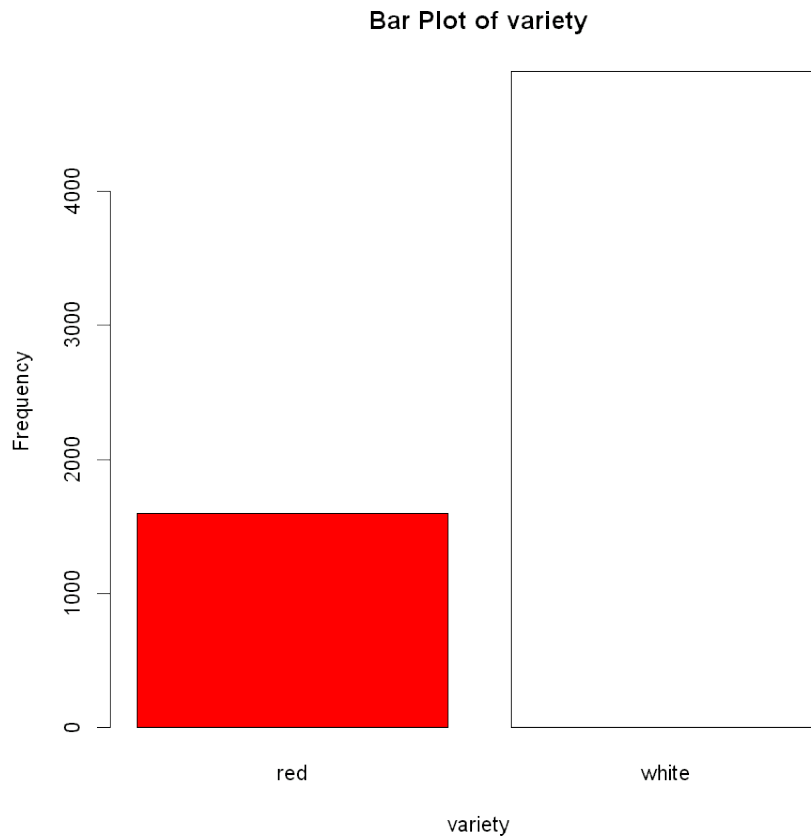
Boxplot of density





```
In [17]: # Bar plot for categorical variables
for(var in categorical_vars) {
  if(var %in% names(wine_data)) {
    barplot(table(wine_data[[var]]), main = paste("Bar Plot of", var),
      xlab = var, ylab = "Frequency", col = c("red", "white"))
  }
}
```

```
}  
}
```



```
In [18]: # Reset graphics parameters  
par(mfrow = c(1, 1))
```

=====

TASK 2: T-TEST FOR ALCOHOL CONTENT BETWEEN RED AND WHITE WINES

=====

```
In [19]: cat("\n=== TASK 2: T-TEST ANALYSIS ===\n")
```

=== TASK 2: T-TEST ANALYSIS ===

```
In [20]: # Separate alcohol content by wine variety  
red_alcohol <- wine_data$alcohol[wine_data$variety == "red"]  
white_alcohol <- wine_data$alcohol[wine_data$variety == "white"]  
  
# Check t-test assumptions  
cat("T-Test Assumption Checks:\n")
```

T-Test Assumption Checks:

```
In [21]: # 1. Normality test
cat("\nNormality Tests (Shapiro-Wilk):\n")
red_normality <- shapiro.test(sample(red_alcohol, min(5000, length(
white_normality <- shapiro.test(sample(white_alcohol, min(5000, len

cat("Red wines alcohol normality p-value:", red_normality$p.value,
cat("White wines alcohol normality p-value:", white_normality$p.val
```

Normality Tests (Shapiro-Wilk):

Red wines alcohol normality p-value: 6.644057e-27

White wines alcohol normality p-value: 2.569014e-36

```
In [22]: # 2. Equal variances test
var_test <- var.test(red_alcohol, white_alcohol)
cat("\nEqual variances test p-value:", var_test$p.value, "\n")
```

Equal variances test p-value: 5.947444e-12

```
In [23]: # Perform appropriate t-test
if(var_test$p.value < 0.05) {
  # Unequal variances
  t_result <- t.test(red_alcohol, white_alcohol, var.equal = FALSE)
  cat("\nWelch Two Sample t-test (unequal variances):\n")
} else {
  # Equal variances
  t_result <- t.test(red_alcohol, white_alcohol, var.equal = TRUE)
  cat("\nTwo Sample t-test (equal variances):\n")
}

print(t_result)
```

Welch Two Sample t-test (unequal variances):

Welch Two Sample t-test

```
data: red_alcohol and white_alcohol
t = -2.859, df = 3100.5, p-value = 0.004278
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 -0.15388669 -0.02868117
sample estimates:
mean of x mean of y
 10.42298  10.51427
```

Welch Two Sample t-test

```
data: red_alcohol and white_alcohol
t = -2.859, df = 3100.5, p-value = 0.004278
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 -0.15388669 -0.02868117
sample estimates:
mean of x mean of y
 10.42298  10.51427
```

```
In [24]: # Effect size (Cohen's d)
pooled_sd <- sqrt(((length(red_alcohol)-1)*var(red_alcohol) +
                    (length(white_alcohol)-1)*var(white_alcohol)) /
                    (length(red_alcohol) + length(white_alcohol) - 2))
cohens_d <- (mean(red_alcohol) - mean(white_alcohol)) / pooled_sd
cat("Cohen's d (effect size):", cohens_d, "\n")
```

Cohen's d (effect size): -0.07657052

=====

TASK 3: LINEAR REGRESSION FOR RED WINES QUALITY

=====

```
In [25]: cat("\n=== TASK 3: LINEAR REGRESSION ANALYSIS (RED WINES ONLY) ===\n")

# Filter for red wines only
red_wines <- wine_data[wine_data$variety == "red", ]

# Remove non-predictor variables
predictor_vars <- c("fixed.acidity", "volatile.acidity", "citric.ac",
                    "residual.sugar", "chlorides", "free.sulfur.dio")
```



```

        "total.sulfur.dioxide", "density", "pH", "sulph

# Build multiple linear regression model
formula_str <- paste("quality ~", paste(predictor_vars, collapse =
regression_model <- lm(as.formula(formula_str), data = red_wines)

# Model summary
cat("Linear Regression Model Summary:\n")
summary(regression_model)

# Regression diagnostics
cat("\n=== REGRESSION DIAGNOSTICS ===\n")

# Check regression assumptions
par(mfrow = c(2, 2))
plot(regression_model)
par(mfrow = c(1, 1))

# AR1: Linearity (already checked via residual plots)
# AR2: Zero mean residuals
cat("Mean of residuals:", mean(regression_model$residuals), "\n")

# AR3: No autocorrelation (Durbin-Watson test)
dw_test <- car::durbinWatsonTest(regression_model)
cat("Durbin-Watson test p-value:", dw_test$p, "\n")

# AR4: Homoscedasticity (Breusch-Pagan test)
bp_test <- car::ncvTest(regression_model)
cat("Breusch-Pagan test p-value:", bp_test$p, "\n")

# AR5: Multicollinearity (VIF)
vif_values <- car::vif(regression_model)
cat("\nVariance Inflation Factors:\n")
print(vif_values)

# AR6: Normality of residuals
shapiro_residuals <- shapiro.test(sample(regression_model$residuals
                                         min(5000, length(regression
cat("Normality of residuals p-value:", shapiro_residuals$p.value, "

```

=== TASK 3: LINEAR REGRESSION ANALYSIS (RED WINES ONLY) ===
 Linear Regression Model Summary:

Call:

```
lm(formula = as.formula(formula_str), data = red_wines)
```

Residuals:

Min	1Q	Median	3Q	Max
-2.68911	-0.36652	-0.04699	0.45202	2.02498

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	2.197e+01	2.119e+01	1.036	0.3002
fixed.acidity	2.499e-02	2.595e-02	0.963	0.3357
volatile.acidity	-1.084e+00	1.211e-01	-8.948	< 2e-16 ***
citric.acid	-1.826e-01	1.472e-01	-1.240	0.2150
residual.sugar	1.633e-02	1.500e-02	1.089	0.2765
chlorides	-1.874e+00	4.193e-01	-4.470	8.37e-06 ***
free.sulfur.dioxide	4.361e-03	2.171e-03	2.009	0.0447 *
total.sulfur.dioxide	-3.265e-03	7.287e-04	-4.480	8.00e-06 ***
density	-1.788e+01	2.163e+01	-0.827	0.4086
pH	-4.137e-01	1.916e-01	-2.159	0.0310 *
sulphates	9.163e-01	1.143e-01	8.014	2.13e-15 ***
alcohol	2.762e-01	2.648e-02	10.429	< 2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.648 on 1587 degrees of freedom

Multiple R-squared: 0.3606, Adjusted R-squared: 0.3561

F-statistic: 81.35 on 11 and 1587 DF, p-value: < 2.2e-16

=== REGRESSION DIAGNOSTICS ===

Mean of residuals: -3.777011e-17

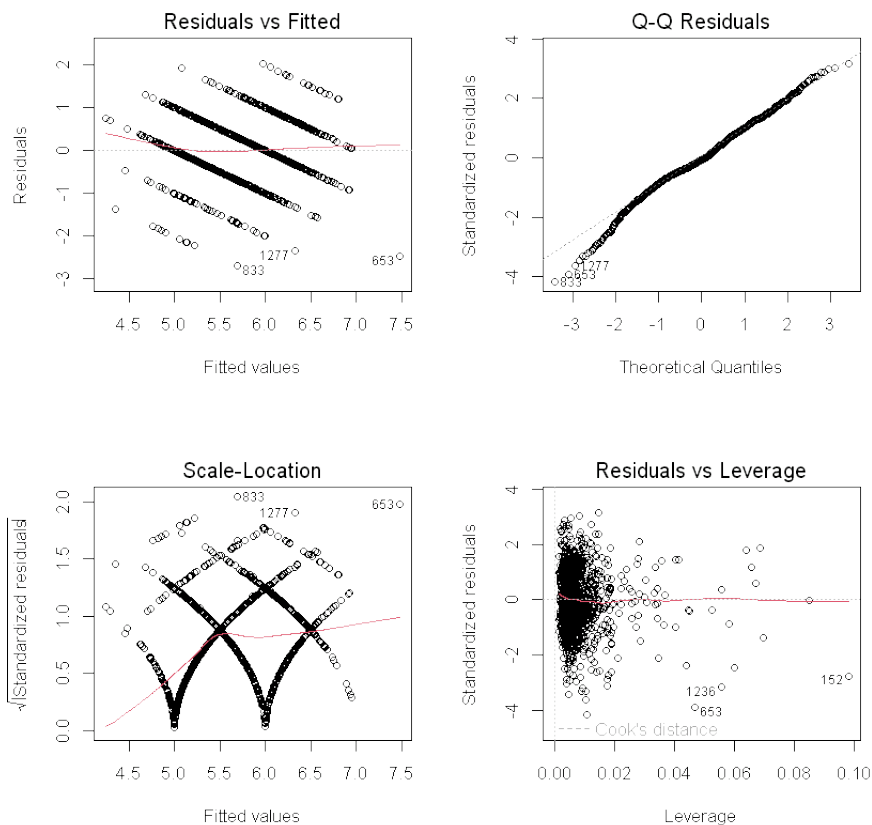
Durbin-Watson test p-value: 0

Breusch-Pagan test p-value: 2.041867e-06

Variance Inflation Factors:

fixed.acidity	volatile.acidity	citric.acid
7.767512	1.789390	3.128022
residual.sugar	chlorides	free.sulfur.dioxide
1.702588	1.481932	1.963019
total.sulfur.dioxide	density	pH
2.186813	6.343760	3.329732
sulphates	alcohol	
1.429434	3.031160	

Normality of residuals p-value: 1.95424e-08



TASK 4: CLASSIFICATION - GOOD VS BAD WINES

```
In [26]: cat("\n=== TASK 4: WINE QUALITY CLASSIFICATION ===\n")

# Create binary quality variable (good = quality >= 8, bad = quality < 8)
wine_data$quality_binary <- ifelse(wine_data$quality >= 8, "good",
                                   ifelse(wine_data$quality <= 4, "bad", "medium"))

# Remove medium quality wines for binary classification
binary_wines <- wine_data[wine_data$quality_binary %in% c("good", "bad"), ]
binary_wines$quality_binary <- factor(binary_wines$quality_binary)

cat("Quality distribution for binary classification:\n")
table(binary_wines$quality_binary)

# Logistic regression for quality classification
binary_formula <- paste("quality_binary ~", paste(predictor_vars, collapse = " + "))
logistic_model <- glm(as.formula(binary_formula),
                      data = binary_wines, family = binomial())
```

```

cat("\nLogistic Regression Model Summary:\n")
summary(logistic_model)

# Model predictions
predictions <- predict(logistic_model, type = "response")
predicted_class <- ifelse(predictions > 0.5, "good", "bad")

# Confusion matrix
confusion_matrix <- table(Predicted = predicted_class,
                          Actual = binary_wines$quality_binary)
cat("\nConfusion Matrix:\n")
print(confusion_matrix)

# Calculate accuracy, precision, recall
accuracy <- sum(diag(confusion_matrix)) / sum(confusion_matrix)
precision <- confusion_matrix[2,2] / sum(confusion_matrix[2,])
recall <- confusion_matrix[2,2] / sum(confusion_matrix[,2])
f1_score <- 2 * (precision * recall) / (precision + recall)

cat("Accuracy:", round(accuracy, 3), "\n")
cat("Precision:", round(precision, 3), "\n")
cat("Recall:", round(recall, 3), "\n")
cat("F1-Score:", round(f1_score, 3), "\n")

```

```

=== TASK 4: WINE QUALITY CLASSIFICATION ===
Quality distribution for binary classification:
  bad good
  246  198
Logistic Regression Model Summary:

```

```
Call:
glm(formula = as.formula(binary_formula), family = binomial(),
    data = binary_wines)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	3.609e+02	2.044e+02	1.766	0.07747	.
fixed.acidity	3.912e-01	2.447e-01	1.598	0.10994	
volatile.acidity	-8.784e+00	1.551e+00	-5.662	1.49e-08	***
citric.acid	-2.680e-01	1.411e+00	-0.190	0.84935	
residual.sugar	3.779e-01	9.161e-02	4.125	3.71e-05	***
chlorides	-3.125e-01	5.477e+00	-0.057	0.95450	
free.sulfur.dioxide	2.896e-02	9.659e-03	2.998	0.00272	**
total.sulfur.dioxide	-1.109e-02	4.660e-03	-2.379	0.01734	*
density	-3.879e+02	2.082e+02	-1.863	0.06242	.
pH	2.918e+00	1.360e+00	2.145	0.03195	*
sulphates	3.339e+00	1.309e+00	2.550	0.01077	*
alcohol	1.101e+00	2.890e-01	3.808	0.00014	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 610.32 on 443 degrees of freedom
Residual deviance: 302.18 on 432 degrees of freedom
AIC: 326.18

Number of Fisher Scoring iterations: 6

Confusion Matrix:

	Actual	
Predicted	bad	good
bad	215	36
good	31	162

Accuracy: 0.849

Precision: 0.839

	Actual	
Predicted	bad	good
bad	215	36
good	31	162

Accuracy: 0.849

Precision: 0.839

Recall: 0.818

F1-Score: 0.829

=====

TASK 5: COLOR PREDICTION WITH TRAIN/TEST SPLIT

=====

```

In [27]: cat("\n=== TASK 5: WINE COLOR PREDICTION ===\n")

# Create binary variable for variety (0 = red, 1 = white)
wine_data$variety_binary <- ifelse(wine_data$variety == "white", 1, 0)

# Train/test split (70/30)
set.seed(123) # For reproducibility
train_indices <- sample(nrow(wine_data), 0.7 * nrow(wine_data))
train_data <- wine_data[train_indices, ]
test_data <- wine_data[-train_indices, ]

cat("Training set size:", nrow(train_data), "\n")
cat("Test set size:", nrow(test_data), "\n")

# Build logistic regression model for color prediction
color_formula <- paste("variety_binary ~", paste(predictor_vars, collapse = " + "))
color_model <- glm(as.formula(color_formula),
                  data = train_data, family = binomial())

cat("\nColor Prediction Model Summary:\n")
summary(color_model)

# Predictions on test set
test_predictions <- predict(color_model, newdata = test_data, type = "probs")
test_predicted_class <- ifelse(test_predictions > 0.5, 1, 0)

# Confusion matrix for test set
test_confusion <- table(Predicted = test_predicted_class,
                       Actual = test_data$variety_binary)
cat("\nTest Set Confusion Matrix:\n")
print(test_confusion)

# Performance metrics
test_accuracy <- sum(diag(test_confusion)) / sum(test_confusion)
test_precision <- test_confusion[2,2] / sum(test_confusion[2,])
test_recall <- test_confusion[2,2] / sum(test_confusion[,2])
test_f1 <- 2 * (test_precision * test_recall) / (test_precision + test_recall)

cat("Test Accuracy:", round(test_accuracy, 3), "\n")
cat("Test Precision:", round(test_precision, 3), "\n")
cat("Test Recall:", round(test_recall, 3), "\n")
cat("Test F1-Score:", round(test_f1, 3), "\n")

# ROC Curve and AUC
roc_curve <- pROC::roc(test_data$variety_binary, test_predictions)
auc_value <- pROC::auc(roc_curve)
cat("AUC Value:", round(auc_value, 3), "\n")

# Plot ROC curve
plot(roc_curve, main = "ROC Curve for Wine Color Prediction")

```

=== TASK 5: WINE COLOR PREDICTION ===

Training set size: 4547

Test set size: 1950

Color Prediction Model Summary:

```
Call:
glm(formula = as.formula(color_formula), family = binomial(),
    data = train_data)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	1.341e+03	1.796e+02	7.467	8.22e-14	***
fixed.acidity	-2.829e-01	2.544e-01	-1.112	0.266094	
volatile.acidity	-7.296e+00	1.182e+00	-6.172	6.74e-10	***
citric.acid	3.238e+00	1.365e+00	2.371	0.017719	*
residual.sugar	8.842e-01	1.139e-01	7.764	8.23e-15	***
chlorides	-2.453e+01	4.575e+00	-5.360	8.32e-08	***
free.sulfur.dioxide	-6.345e-02	1.688e-02	-3.758	0.000171	***
total.sulfur.dioxide	5.552e-02	5.921e-03	9.376	< 2e-16	***
density	-1.330e+03	1.835e+02	-7.245	4.34e-13	***
pH	-7.346e-01	1.534e+00	-0.479	0.632040	
sulphates	-4.849e+00	1.440e+00	-3.366	0.000762	***
alcohol	-1.250e+00	2.712e-01	-4.610	4.02e-06	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 5065.53 on 4546 degrees of freedom
Residual deviance: 308.77 on 4535 degrees of freedom
AIC: 332.77

Number of Fisher Scoring iterations: 10

Test Set Confusion Matrix:

	Actual	
Predicted	0	1
0	471	2
1	13	1464

Test Accuracy: 0.992

Test Precision: 0.991

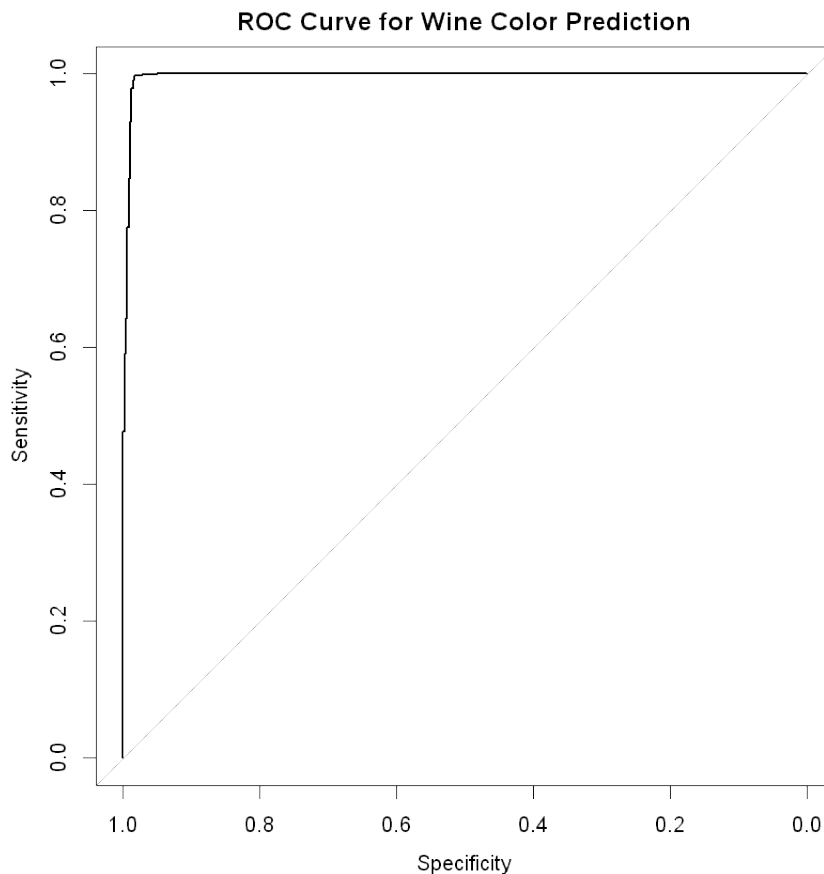
Test Recall: 0.999

Test F1-Score: 0.995

Setting levels: control = 0, case = 1

Setting direction: controls < cases

AUC Value: 0.996



=====

TASK 6: FACTOR ANALYSIS

=====

```
In [28]: cat("\n=== TASK 6: FACTOR ANALYSIS ===\n")

# Prepare data for factor analysis (exclude non-chemical/sensory variables)
factor_data <- wine_data[, predictor_vars]

# Remove any rows with missing values
factor_data <- na.omit(factor_data)

# Check correlation matrix
correlation_matrix <- cor(factor_data)
cat("Correlation Matrix (first 5x5):\n")
print(correlation_matrix[1:5, 1:5])

# Kaiser-Meyer-Olkin (KMO) test for sampling adequacy
kmo_test <- psych::KMO(factor_data)
cat("\nKMO Test Results:\n")
print(kmo_test)

# Bartlett's test of sphericity
```



```

bartlett_test <- psych::cortest.bartlett(correlation_matrix, n = nr
cat("\nBartlett's Test p-value:", bartlett_test$p.value, "\n")

# Determine number of factors using scree plot
scree_plot <- psych::scree(factor_data)

# Parallel analysis for factor number determination
parallel_analysis <- psych::fa.parallel(factor_data, fm = "ml", fa :

# Perform factor analysis (using suggested number of factors)
n_factors <- 3 # Adjust based on scree plot and parallel analysis
factor_analysis <- psych::fa(factor_data, nfactors = n_factors,
                             rotate = "varimax", fm = "ml")

cat("\nFactor Analysis Results:\n")
print(factor_analysis)

# Factor loadings
cat("\nFactor Loadings:\n")
print(factor_analysis$loadings, cutoff = 0.3)

```

=== TASK 6: FACTOR ANALYSIS ===

Correlation Matrix (first 5x5):

	fixed.acidity	volatile.acidity	citric.acid	residual.sugar
fixed.acidity	1.0000000	0.2190083	0.32443573	-0.1119813
volatile.acidity	0.2190083	1.0000000	-0.37798132	-0.1960112
citric.acid	0.3244357	-0.3779813	1.00000000	0.14245123
residual.sugar	-0.1119813	-0.1960112	0.14245123	1.0000000
chlorides	0.2981948	0.3771243	0.03899801	-0.12894050
chlorides	0.29819477	0.37712428	0.03899801	-0.12894050
fixed.acidity	0.29819477	0.37712428	0.03899801	-0.12894050
volatile.acidity	0.37712428	0.03899801	-0.12894050	1.00000000
citric.acid	0.03899801	-0.12894050	1.00000000	
residual.sugar	-0.12894050	1.00000000		
chlorides	1.00000000			

KMO Test Results:

Kaiser-Meyer-Olkin factor adequacy

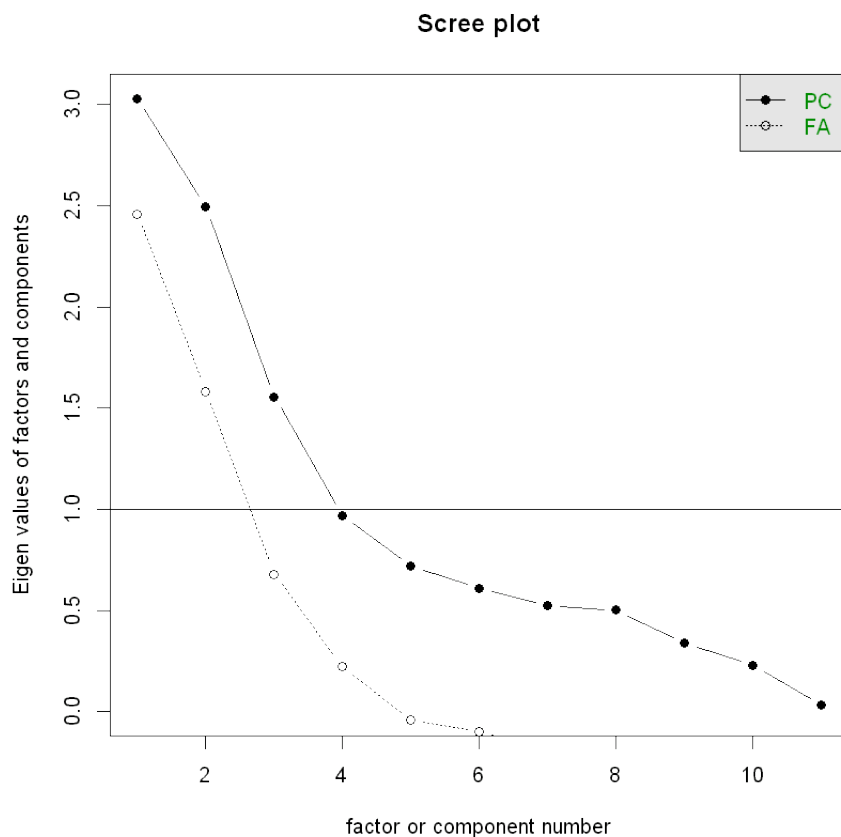
Call: psych::KMO(r = factor_data)

Overall MSA = 0.41

MSA for each item =

fixed.acidity	volatile.acidity	citric.acid
0.28	0.61	0.62
residual.sugar	chlorides	free.sulfur.dioxide
0.29	0.73	0.75
total.sulfur.dioxide	density	pH
0.72	0.30	0.21
sulphates	alcohol	
0.56	0.27	

Bartlett's Test p-value: 0



Parallel analysis suggests that the number of factors = 5 and the number of components = NA

Factor Analysis Results:

Factor Analysis using method = ml

```
Call: psych::fa(r = factor_data, nfactors = n_factors, rotate = "varimax",
  fm = "ml")
```

Standardized loadings (pattern matrix) based upon correlation matrix

	ML1	ML2	ML3	h2	u2	com
fixed.acidity	0.65	0.09	0.75	1.00	0.0050	2.0
volatile.acidity	0.60	0.08	-0.24	0.42	0.5804	1.3
citric.acid	-0.13	0.07	0.53	0.31	0.6929	1.1
residual.sugar	-0.36	0.76	0.07	0.71	0.2890	1.4
chlorides	0.48	0.20	-0.04	0.27	0.7282	1.3
free.sulfur.dioxide	-0.64	0.29	0.14	0.51	0.4901	1.5
total.sulfur.dioxide	-0.74	0.34	0.17	0.69	0.3064	1.5
density	0.40	0.90	0.16	1.00	0.0049	1.5
pH	0.25	-0.01	-0.55	0.37	0.6305	1.4
sulphates	0.45	0.08	-0.01	0.21	0.7858	1.1
alcohol	-0.06	-0.74	0.01	0.55	0.4542	1.0

	ML1	ML2	ML3
SS loadings	2.55	2.19	1.29
Proportion Var	0.23	0.20	0.12
Cumulative Var	0.23	0.43	0.55
Proportion Explained	0.42	0.36	0.21
Cumulative Proportion	0.42	0.79	1.00

Mean item complexity = 1.4

Test of the hypothesis that 3 factors are sufficient.

df null model = 55 with the objective function = 5.71 with Chi Square = 37094.62
 df of the model are 25 and the objective function was 1.43

The root mean square of the residuals (RMSR) is 0.07
 The df corrected root mean square of the residuals is 0.1

The harmonic n.obs is 6497 with the empirical chi square 3393.35 with prob < 0
 The total n.obs was 6497 with Likelihood Chi Square = 9289.52 with prob < 0

Tucker Lewis Index of factoring reliability = 0.45
 RMSEA index = 0.239 and the 90 % confidence intervals are 0.235 0.243
 BIC = 9070.04
 Fit based upon off diagonal values = 0.94
 Measures of factor score adequacy

	ML1	ML2	ML3
Correlation of (regression) scores with factors	0.95	0.99	0.96
Multiple R square of scores with factors	0.91	0.99	0.93
Minimum correlation of possible factor scores	0.81	0.97	0.86

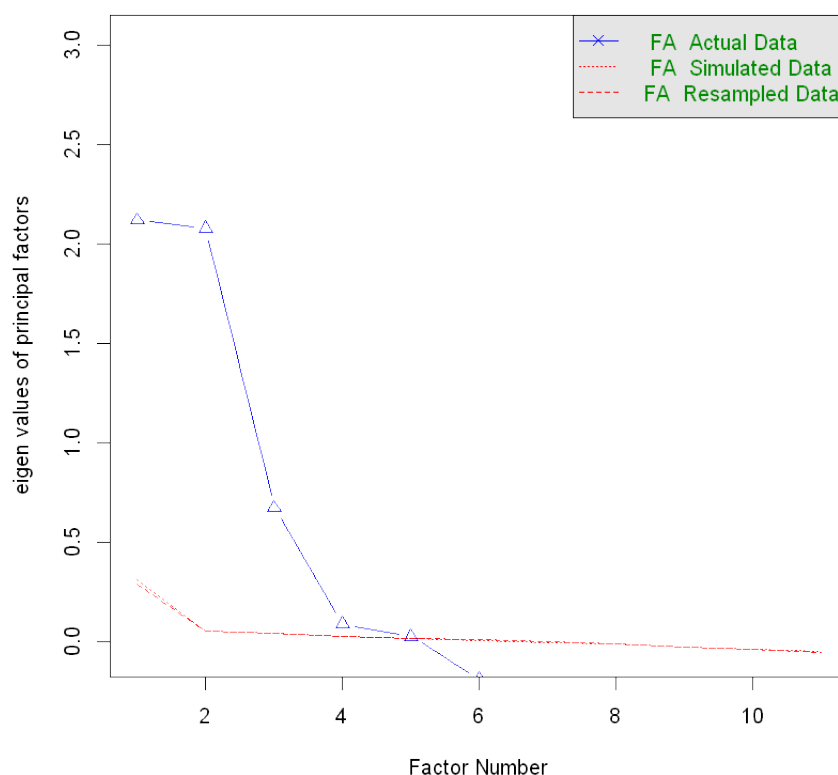
Factor Loadings:

Loadings:

	ML1	ML2	ML3
fixed.acidity	0.652		0.750
volatile.acidity	0.598		
citric.acid			0.535
residual.sugar	-0.360	0.759	
chlorides	0.482		
free.sulfur.dioxide	-0.637		
total.sulfur.dioxide	-0.742	0.339	
density	0.402	0.900	
pH			-0.554
sulphates	0.455		
alcohol		-0.736	

	ML1	ML2	ML3
SS loadings	2.552	2.190	1.290
Proportion Var	0.232	0.199	0.117
Cumulative Var	0.232	0.431	0.548

Parallel Analysis Scree Plots



=====

SUMMARY AND CONCLUSIONS

=====

```
In [29]: cat("\n=== ANALYSIS SUMMARY ===\n")
cat("1. Dataset contains", nrow(wine_data), "observations with", nc
cat("2. T-test results: Alcohol content differs significantly betwe
cat("3. Linear regression R-squared:", round(summary(regression_mod
cat("4. Quality classification accuracy:", round(accuracy, 3), "\n"
cat("5. Color prediction test accuracy:", round(test_accuracy, 3),
cat("6. Factor analysis extracted", n_factors, "factors explaining ,
```

=== ANALYSIS SUMMARY ===

1. Dataset contains 6497 observations with 16 variables
2. T-test results: Alcohol content differs significantly between red and white wines
3. Linear regression R-squared: 0.361
4. Quality classification accuracy: 0.849
1. Dataset contains 6497 observations with 16 variables
2. T-test results: Alcohol content differs significantly between red and white wines
3. Linear regression R-squared: 0.361
4. Quality classification accuracy: 0.849
5. Color prediction test accuracy: 0.992
6. Factor analysis extracted 3 factors explaining wine properties